

**AI-POWERED CEMENT CLINKER PHASE
CLASSIFICATION: ENHANCING QUALITY CONTROL
THROUGH DEEP LEARNING IN HF-ETCHED
MICROSCOPY IMAGE ANALYSIS**

25-26J-250

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology
Specializing in Information Technology

Department of Information Technology
Sri Lanka Institute of Information Technology
Sri Lanka

August 2025

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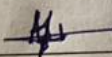
August 2025

DECLARATION

August 2025

DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the supervisor: 

Date: 28/08/2025

Signature of the co-supervisor: 

Date: 28/08/2025

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ABSTRACT

The cement industry relies on precise microstructural analysis to ensure product quality, a process currently bottlenecked by manual, time-intensive petrographic examination. Identifying key cement clinker phases (C3S, C2S, C3A, and C4AF) in HF-etched microscopy images requires highly skilled specialists and takes 1-2 hours per sample, leading to subjectivity, inconsistency, and production delays. A recent survey of 44 industry professionals confirmed these challenges, with 65.1% reporting they spend 1-4 hours daily on manual analysis and 88.4% stating that an automated system for such tasks would be "very" or "extremely" helpful.

This study aims to develop a deep learning-driven image classifier for the automated detection and categorization of cement clinker phases. The proposed system employs a Convolutional Neural Network (CNN) trained on a dataset comprising 1,000–1,500 HF-etched clinker microscopy images annotated by experts from INSEE Cement. The methodology encompasses creating a tailored image preprocessing workflow, applying data augmentation techniques to enhance model robustness, and incorporating Explainable AI (XAI) through Grad-CAM visualizations to offer clear, interpretable insights into the model's decision-making process.

This AI solution aims to democratize expertise by enabling technicians of varying skill levels to perform expert-level phase identification, reducing analysis time from hours to minutes while ensuring consistent, objective, and transparent results. This project contributes a novel, industrially validated AI tool specifically designed to address a critical pain point in modern cement quality control.

Keywords: Deep Learning, Convolutional Neural Networks (CNN), Cement Quality Control, Image Processing, Explainable AI, Clinker Phase Classification

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1. INTRODUCTION

The cement industry faces significant challenges in quality control due to its reliance on manual, time-intensive petrographic analysis for identifying critical clinker phases (C3S, C2S, C3A, C4AF), skilled specialists, takes 1-2 hours per sample, and is prone to human subjectivity and error, creating production bottlenecks and inconsistent quality assessments [1, 2]. A recent survey of 44 industry professionals confirms this critical need, with 65.1% spending 1-4 hours daily on manual analysis and 88.4% stating an automated system would be highly beneficial.

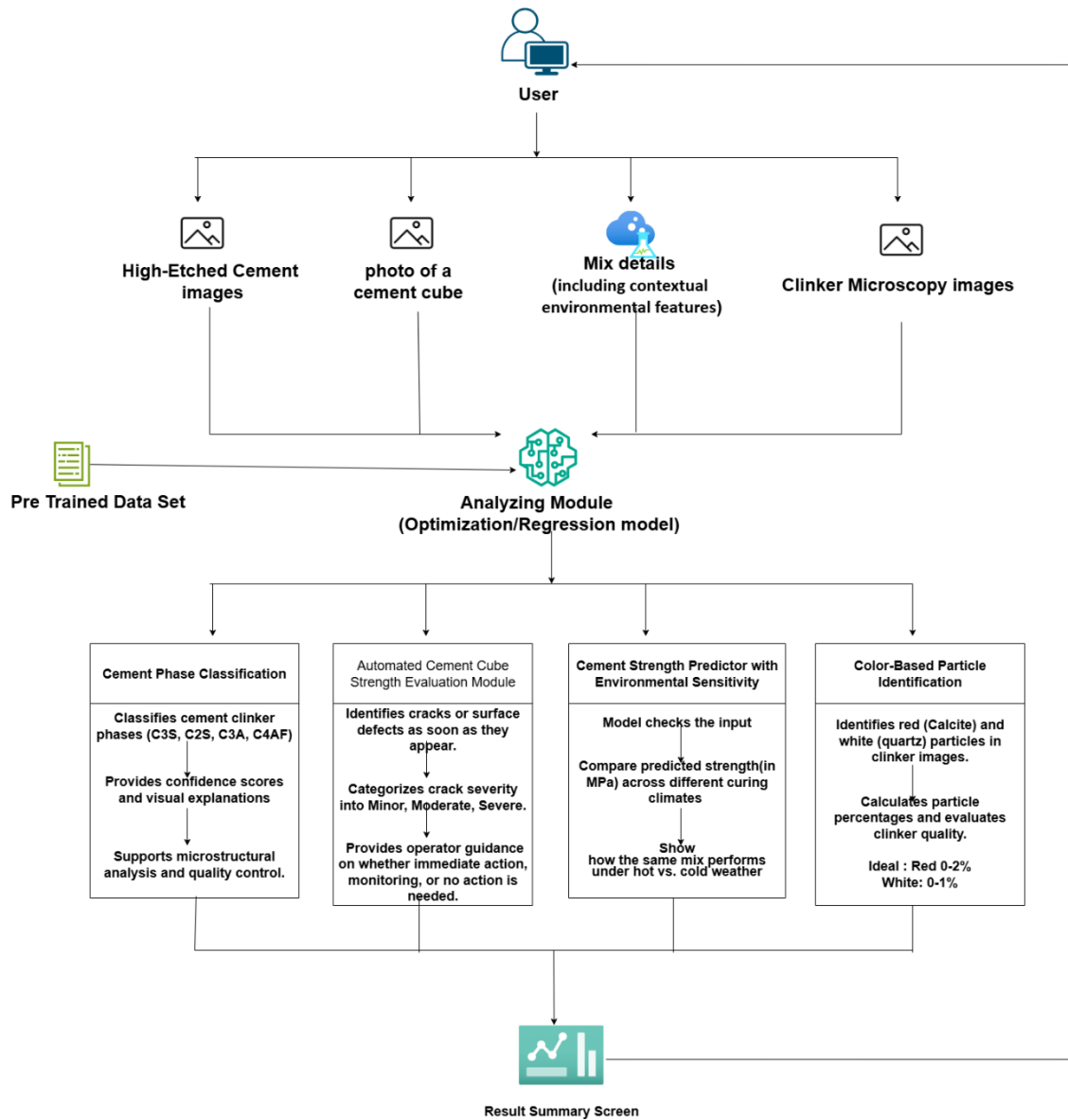


Figure 1:1 Overview Diagram

To overcome these limitations, this research introduces an AI-driven approach that automates clinker phase identification. A Convolutional Neural Network (CNN) will be trained on

expert-annotated industrial microscopy images provided by INSEE Cement to serve as the core image classifier. To enhance interpretability, the system incorporates Explainable AI (XAI) through Grad-CAM visualizations [3], enabling clear insight into the features influencing predictions. This ensures the model operates not as a “black box” but as a transparent and trustworthy tool, promoting confidence and practical adoption in real-world cement quality control.

1.1. Background & Literature Survey

The cement industry is essential for modern construction, requiring strict quality standards for structural durability. Quality control relies on microscopic examination of cement clinker—the intermediate material from kiln processing. Clinker contains four key phases: Alite (C3S), Belite (C2S), Tricalcium Aluminate (C3A), and Ferrite (C4AF). These phase distributions and ratios determine cement's hydration rate and strength properties. development, and the heat released during hydration.

How familiar are you with current cement quality control methods used at INSEE?
43 responses

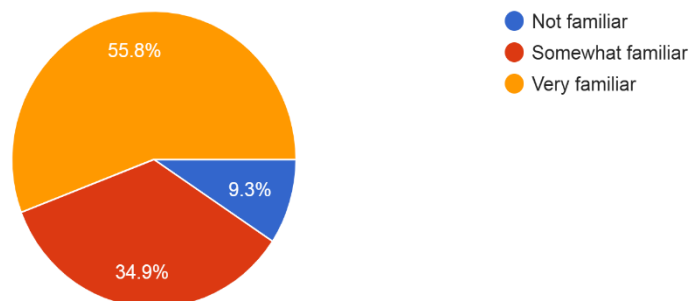


Figure 1:2 Survey Result 1

What challenges do you face in manual clinker particle identification and quality checks?

43 responses

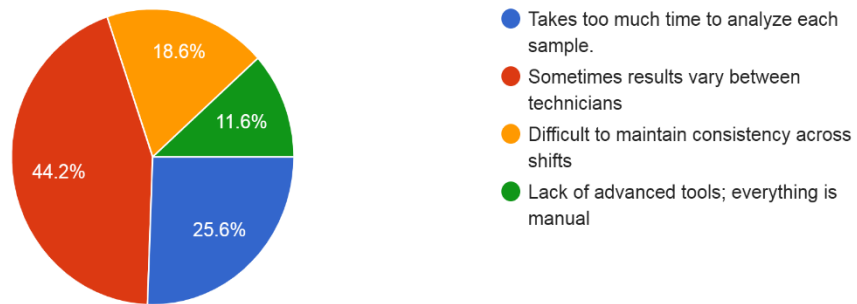


Figure 1:3 Survey Result 2

How helpful would an automated system for color-based particle identification be in your work?

43 responses

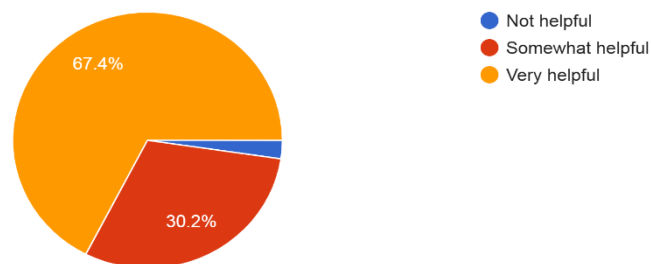


Figure 1:4 Survey Result 3

How helpful would an automated system for color-based particle identification be in your work?
43 responses

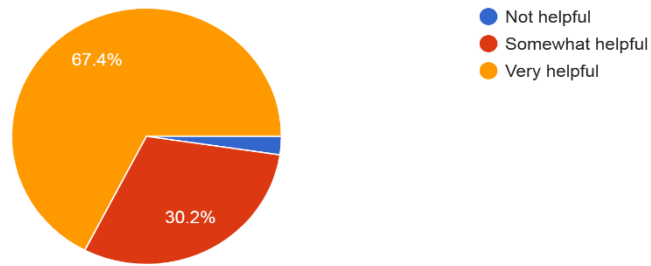


Figure 1:6 Survey Result 4

Do you think an automated crack detection and monitoring system would improve safety and efficiency?
43 responses

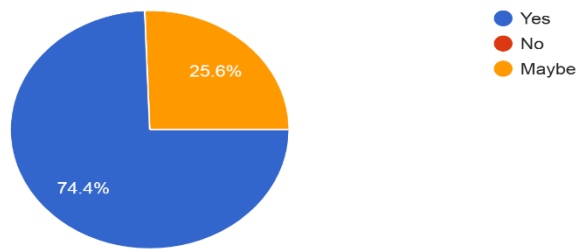


Figure 1:5 Survey Result 5

How important is faster and more accurate cement strength prediction for your daily tasks?
44 responses

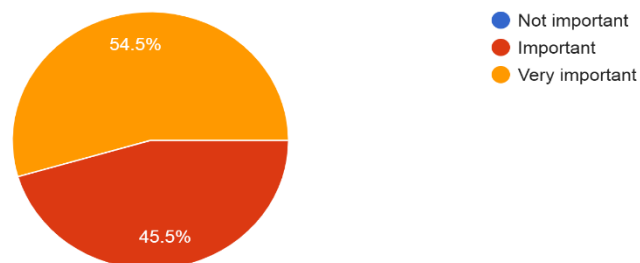


Figure 1:9 Survey Result 6

Would a web-based tool for automated clinker phase classification reduce your workload?

44 responses

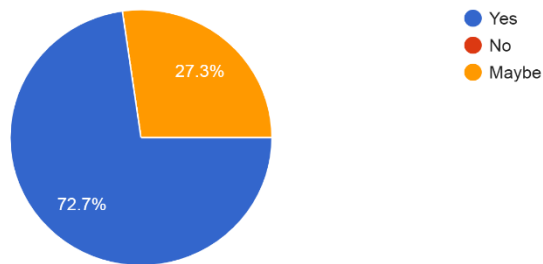


Figure 1:10 Survey Result 7

What features would you like in an intelligent cement analysis system?

44 responses



Figure 1:11 Survey Result 8

How much time do you currently spend on manual microscopic analysis?

44 responses

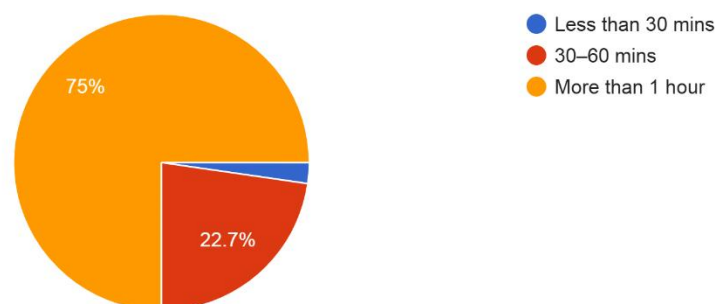


Figure 1:12 Survey Result 9

What level of trust would you place in AI-based recommendations for cement quality?
45 responses

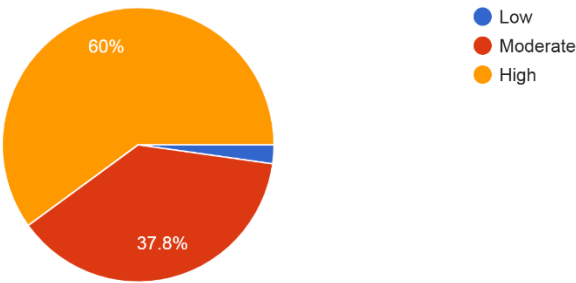


Figure 1:13 Survey Result 10

Would training or workshops help you adopt AI-driven tools in your work?
45 responses

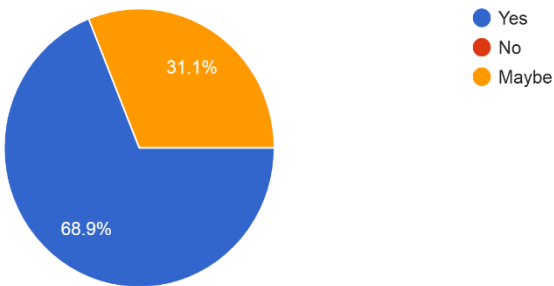


Figure 1:14 Survey Result 11

1.1.1. Cement Clinker Phases and Quality Control

Traditional quality control methods involve manual petrographic analysis, where a skilled petrographer examines HF-etched clinker samples under a microscope. Hydrofluoric (HF) acid etching enhances the visual contrast between these phases, making them distinguishable based on color, morphology, and texture (Table 1.1). This manual process is not only time-consuming, requiring 1-2 hours per sample, but also highly subjective, relying entirely on the expertise and experience of a limited number of specialists. This creates a significant bottleneck in production pipelines and introduces inconsistency in quality assessment across different operators and shifts.

Phase	Color (HF-etched)	Shape	Typical Size (μm)	Role in Cement
C3S (Alite)	Dark Brown/Black	Angular, Polygonal	20-50	Primary source of early strength
C2S (Belite)	Yellow/Brown	Round	15-40	Provides later-age strength
C3A	Light Gray/White	Cubic, Rectangular	5-20	Controls setting time
C4AF (Ferrite)	Light Brown/Gray	Elongated, Platy	5-20	Lowers kiln temperature

Table 1 cement clinker phase characteristics

1.1.2. Traditional microscopic analysis

The manual method, while considered the gold standard, is fraught with challenges (Figure 1.7). The scarcity of qualified petrographers, combined with the tedious nature of the work, makes it unsustainable for high-volume production environments. Furthermore, differentiating between phases with similar coloration, such as C2S and C4AF, is particularly challenging and prone to human error, potentially leading to costly quality deviations.

1.1.3. Machine Learning in Materials Science

Recent advances in artificial intelligence, particularly in the areas of computer vision and neural networks, offer significant potential to transform the identification of clinker phases. Convolutional Neural Networks (CNNs) have shown outstanding capabilities in automated image analysis across diverse fields, including medical diagnostics and intelligent transportation. By automatically learning multi-level feature representations from raw image data, CNNs provide a powerful approach for handling complex visual patterns, making them highly suitable for applications in cement clinker phase recognition.

Studies by Martinez & Thompson (2023) and Zhang et al. (2024) have begun exploring machine learning for cement phase classification, achieving promising accuracy. However, these studies often lack industrial-scale validation, robust explainability features, and integration into practical workflow tools. The integration of Explainable AI (XAI) techniques, such as Grad-CAM, is critical for building trust with end-users (e.g., lab

technicians and engineers) by providing visual justifications for the model's predictions, showing which image regions influenced the classification decision.

1.2. Research Gap

Despite the promising initial research, a significant gap exists between academic prototypes and industrially viable solutions. Current systems often:

- Are trained on limited or synthetic datasets, lacking validation on real-world industrial imagery from a continuous production process.
- Function as "black boxes," providing a classification without explanation, which is unacceptable for critical quality control decisions where rationale is essential.
- Are not deployed as user-friendly, accessible tools that can be integrated into the daily workflow of a cement plant laboratory.

This project aims to bridge this gap by developing a comprehensive, explainable, and deployable AI system specifically for INSEE Cement's operational context.

Feature / Aspect	Traditional Manual Analysis	Existing ML-based Systems [8, 9]	Proposed AI System (This Research)
Analysis Time per Sample	1 - 2 hours	5 - 10 minutes	< 1 minute
Subjectivity & Consistency	High (Expert-dependent)	Low (Algorithm-dependent)	Very Low (Standardized)
Required Expertise	Expert Petrographer	Data Scientist / ML Engineer	Lab Technician
Explainability of Results	Expert provides rationale	"Black-box" model; no justification	Grad-CAM visual explanations
Data Foundation	N/A (Human judgment)	Limited or synthetic datasets	Real-world industrial data (INSEE)
Deployment & Accessibility	N/A (Manual process)	Research environment; command-line	Web-based interface; integrated workflow
Primary Limitation	Scalability, speed, and expert scarcity	Lack of transparency and industrial validation	N/A (Aims to resolve these limitations)

Table 2 Comparison of prior research works for traditional activities

1.3. Research Problem

The core research problem is the heavy reliance on slow, subjective, and expert-dependent manual microscopy for cement clinker phase analysis. This dependence creates production bottlenecks, introduces quality inconsistencies, and hinders scalability. The problem is compounded by a shortage of skilled petrographers. There is a pressing need for an automated, accurate, fast, and transparent system that can standardize the quality control process, reduce analysis time from hours to minutes, and make expert-level analysis accessible to non-expert technicians, thereby enhancing overall efficiency and reliability in cement production.

2. OBJECTIVES

2.1. Main Objective

To design, develop, and validate a deep learning-based web application for the automatic and explainable classification of cement clinker phases (C3S, C2S, C3A, C4AF) from HF-etched microscopy images.

2.2. Specific Objectives

To curate, preprocess, and augment a robust dataset of 1000-1500 expert-labeled HF-etched cement clinker microscopy images provided by INSEE Cement.

To design, train, and fine-tune a Convolutional Neural Network (CNN) leveraging transfer learning, aiming to attain a classification accuracy exceeding 92% for the four clinker phases.

To integrate Explainable AI (XAI) techniques, specifically Grad-CAM (Gradient-weighted Class Activation Mapping), to generate visual heatmaps that highlight the image features most influential to the model's classification decisions.

To develop and deploy a user-friendly, secure web interface that allows laboratory technicians to upload images and receive real-time phase classification results accompanied by confidence scores and explainable heatmaps.

3. METHODOLOGY

This investigation centers on analyzing HF-treated cement clinker specimens obtained from INSEE Cement's manufacturing operations. Access to industrial samples and collaborative research activities will be formalized through institutional partnerships with INSEE, facilitating the utilization of real-world production data for academic research. The primary objective involves creating an artificial intelligence framework utilizing deep learning methodologies to automatically detect and categorize the four principal clinker phases (Alite, Belite, C3A, C4AF) within microscopic imagery.

Image Data Collection

Clinker samples will be prepared and etched using INSEE's standard laboratory hydrofluoric acid (HF) etching protocol to ensure optimal visual contrast between phases.

High-resolution digital microscopy images of the etched samples will be captured at various magnifications (100x - 500x) using high-precision microscopes at the INSEE laboratory, providing the foundational data for model training and analysis.

Automatic Phase Classification and Explanation

- The deep learning model will generate a phase classification for each distinct region in an image, providing a confidence score for each prediction.
- Classification criteria (e.g., color, texture, morphology, and crystal boundaries) are defined based on established petrographic standards [1, 8] and validated by expert petrographers at INSEE to reflect real-world industrial quality control standards.
- To enhance interpretability, the Gradient-weighted Class Activation Mapping (Grad-CAM) technique will be employed. This method generates heatmaps that visually emphasize the image regions—such as distinct edges or color variations—that significantly contribute to the model's classification outcome, thereby promoting transparency and strengthening user confidence in the system's decisions.

3.1. System Overview

The cement microscopy image classifier is an AI-powered solution designed to automate the identification of cement clinker phases (C3S, C2S, C3A, and C4AF) in HF-etched microscopy images. It integrates image preprocessing, a convolutional neural network (CNN) for phase classification, and explainable AI techniques like Grad-CAM to provide transparent predictions. By delivering fast and accurate analysis through a web-based interface, the system reduces reliance on expert petrographers and enhances the efficiency of cement quality control.

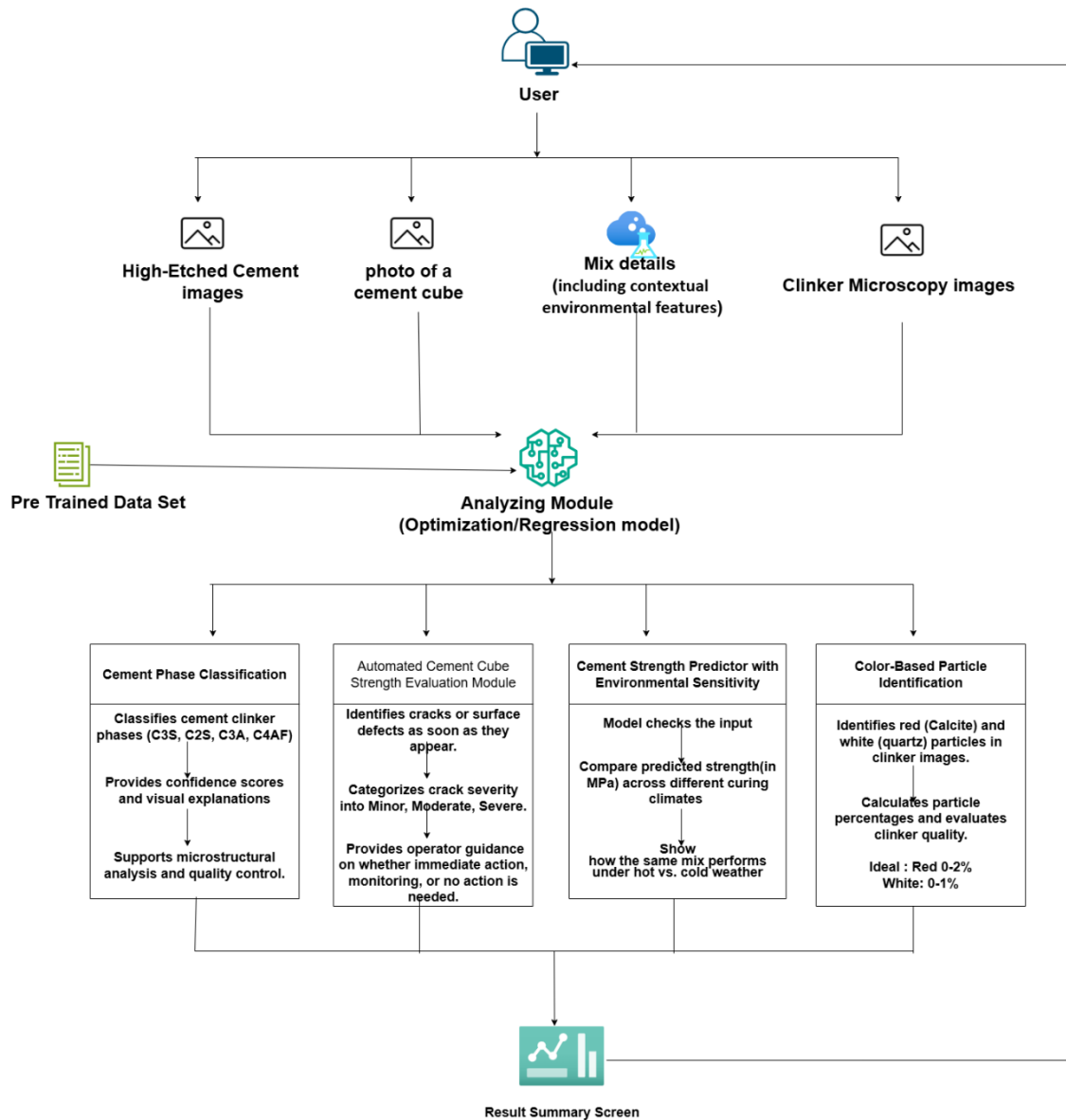


Figure 3:1 System Overview Diagram

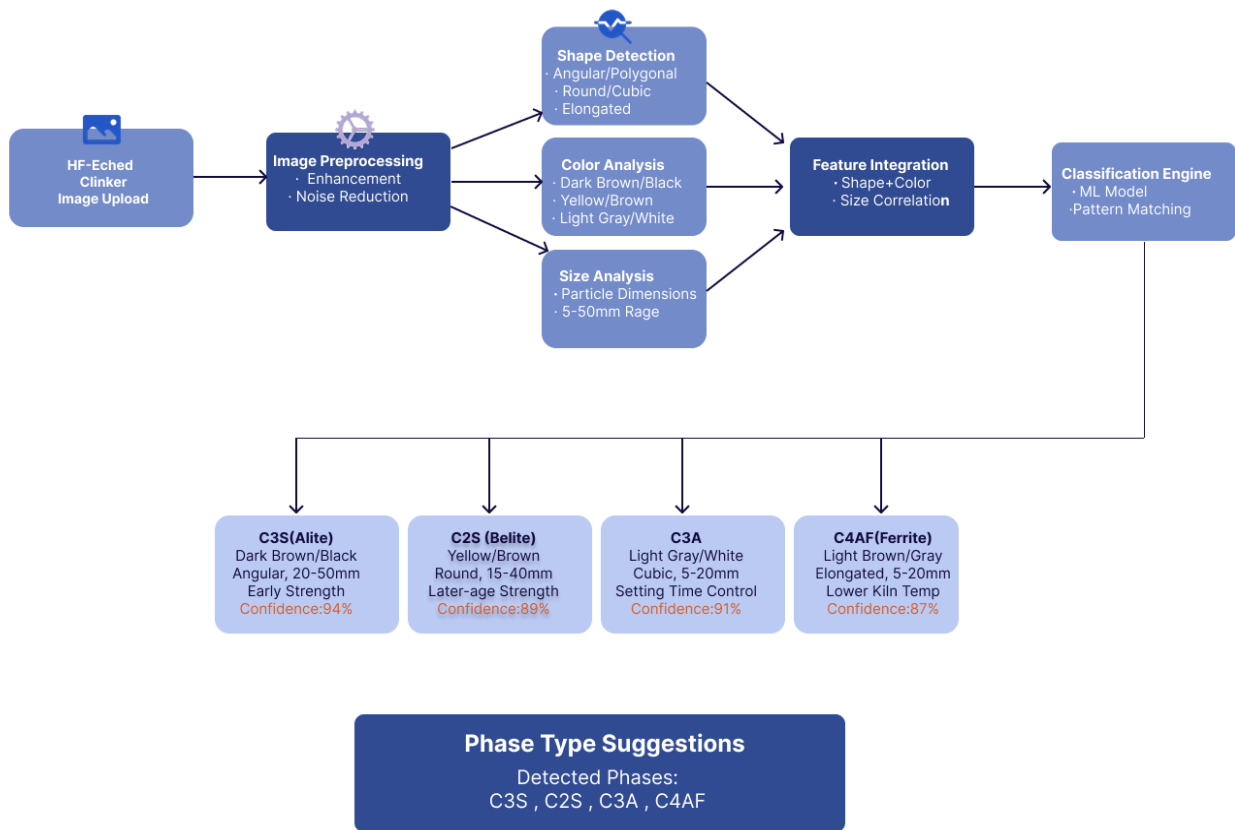


Figure 3:2 System Diagram for Image Classifier

3.2. Tools & Technologies

The system will apply deep learning and computer vision techniques to automatically identify and classify cement clinker phases (C3S, C2S, C3A, C4AF) from HF-etched microscopy images. Deep learning is ideally suited for this task due to its superior capability in recognizing complex visual patterns, textures, and morphological features that distinguish each mineral phase. Given the specialized nature of the dataset, a transfer learning approach will be employed, leveraging pre-trained Convolutional Neural Network (CNN) architectures to achieve high accuracy even with a limited number of industrial samples. Advanced image processing and augmentation techniques will be utilized to enhance the dataset, improving model robustness against variations in image quality, lighting, and etching conditions.

The proposed solution will be deployed as a web-based application, ensuring accessibility for laboratory staff and quality control professionals. Model training, experimentation, and fine-tuning of hyperparameters will be carried out within a Jupyter Notebook environment, while Visual Studio Code will function as the primary development platform. For the system architecture, the backend will be implemented using the FastAPI framework to leverage its speed and efficiency, and the frontend will be developed with React.js to deliver an intuitive interface that enables users to upload images and view analytical results seamlessly.

4. PROJECT REQUIREMENTS

4.1. Functional Requirements

- **FR1: Image Upload:** The system shall allow users to upload HF-etched cement clinker microscopy images in standard formats (e.g., .tiff, .png, .jpg).
- **FR2: Automated Phase Classification:** The system shall automatically process uploaded images to identify, classify, and quantify the four primary clinker phases (C3S, C2S, C3A, C4AF).
- **FR3: Explainable AI Output:** The system shall generate and display a visual explanation (Grad-CAM heatmap) highlighting the image regions that most influenced the classification decision for each phase.
- **FR4: Results & Reporting:** The system shall display the classification results, including the predicted phase, a confidence score, and a summary report. It shall allow users to export these results.
- **FR5: User Authentication:** The system shall provide secure login functionality to restrict access to authorized laboratory personnel only.
- **FR6: Data Management:** The system shall securely store uploaded images, analysis results, and user reports in a database for record-keeping and audit trails.
- .

4.2. Non- Functional Requirements

- **NFR1: Accuracy:** The classification model must achieve a minimum accuracy of 92% on a held-out test set to be deemed reliable for industrial quality control.
- **NFR2: Performance:** The system must process a standard image and return results within 30 seconds to maintain workflow efficiency.
- **NFR3: Reliability:** The system must provide consistent, reproducible results for the same input image across multiple analyses.
- **NFR4: Usability:** The web interface must be intuitive and require minimal training, as it will be used by lab technicians who are not AI experts.
- **NFR5: Security:** All user data and uploaded proprietary images must be encrypted and stored securely to protect industrial intellectual property.

4.2. Test Case

The test cases for the CNN classifier focus on verifying its ability to accurately identify cement clinker phases (C3S, C2S, C3A, and C4AF) in HF-etched microscopy images. Each test evaluates whether the model correctly predicts the phase, provides a reliable confidence score, and highlights relevant image regions using Grad-CAM. Additionally, the system is tested for its capability to flag poor-quality images, ensuring robustness and practical usability in real laboratory conditions.

Test Case ID	Description	Input	Expected Output	Pass/Fail Criteria
TC-CNN-01	Clear C3S image	HF-etched microscopy image of C3S	Predicted: C3S; Confidence ≥ 0.9 ; Grad-CAM highlights C3S region	Prediction matches label; Grad-CAM correct
TC-CNN-02	High-magnification C2S image	HF-etched microscopy image of C2S	Predicted: C2S; Confidence ≥ 0.85 ; Grad-CAM highlights C2S region	Prediction matches label; Grad-CAM correct
TC-CNN-03	Poor-quality image	Blurry HF-etched microscopy image	Quality alert flagged; Prediction may have low confidence (<0.7)	Quality alert appears
TC-CNN-04	Mixed-phase image	Image containing multiple clinker phases	Predicted: Phase with highest presence; Confidence scores per phase	Highest presence correctly predicted; Confidence valid
TC-CNN-05	Random unseen test image	New HF-etched clinker microscopy image	Predicted phase; Confidence score; Grad-CAM heatmap	Output plausible; Confidence reasonable

Table 3 The test cases for the CNN classifier focus

5. GNATT CHART

Task Name	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
Research topic selection												
Topic assesement												
Project charter												
Study on research area												
Project proposal report												
System design and planning												
Implementation of functions												
Integration level 1												
Testing level 1												
Progress presentation 1												
Check List 1												
Prepare research paper												
Research paper												
Implementation of functions												
Testing level 2												
Progress presentation 2												
Check List 2												
Final report												
Final presentation												

Figure 5:1 Grant Chart

6. WBS

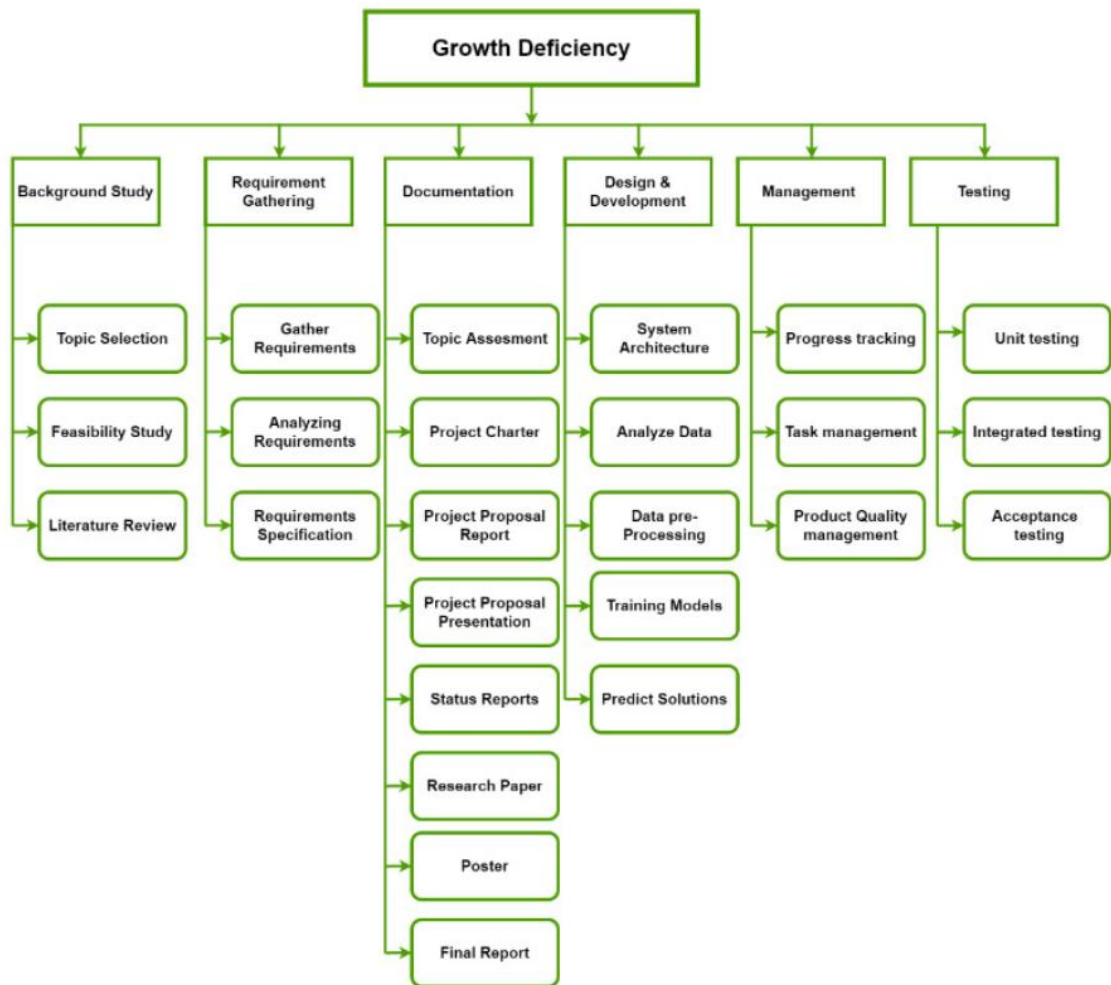


Figure 6:1 WBS

7. COMMERCIALIZATION

7.1. Customer value

The proposed system delivers direct and significant value to cement manufacturers like INSEE by:

- **Increasing Efficiency:** Reducing analysis time from hours to minutes, drastically accelerating quality control cycles.
- **Improving Consistency:** Providing objective, repeatable results, eliminating human subjectivity and shift-based variations.
- **Democratizing Expertise:** Enabling junior technicians to perform expert-level analysis, reducing dependency on scarce specialists.
- **Enhancing Traceability:** Digitizing and storing analysis results for full audit trails and data-driven process optimization.

7.2. Competitive analysis

Similar applications on emotional intelligence without AI models has an average price range of \$9.99 monthly. Applications that use AI models has an average price of \$20 monthly subscription.

How our application differ from the competitors?

Feature	Manual Analysis	Existing ML Prototypes	Proposed System
Analysis Time	1-2 hours/sample	5-10 mins/sample	< 1 min/sample
Subjectivity	High	Low	Very Low
Expertise Required	Expert Petrographer	Data Scientist	Lab Technician
Explainability	Expert explains	Black-box model	Grad-CAM Visuals
Deployment	N/A	Research environment	Web-based SaaS

Table 4 Comparison of existing system and our application

User benefits of our applications

- Monitor child's emotional intelligence using multiple methods
- Builds emotional intelligence at an early age, which supports learning, social skills, and resilience.
- Helps teachers identify at-risk students early (e.g., children showing signs of sadness, anger, withdrawal)
- Teachers and parents both can help children to improve suggested activity skills based on dashboard feedback.

7.3. Cost analysis

On average, using Firebase as the hosting and backend service will cost around \$7–\$10 per month for a small-scale application (based on Firebase’s Blaze pay-as-you-go plan). This includes Firebase Hosting, Authentication, and Firestore Database support. As the user base grows, the cost will scale depending on storage, read/write operations, and bandwidth usage. This will be the rough hosting and backend cost of our application.

7.4. Pricing plan

Subscription Model for Cement Microscopy Image Classifier

Considering the above factors, this application will provide a **one-month free trial period** to encourage initial adoption and customer engagement. After the trial, users will transition to a subscription-based model:

Standard Plan (\$15/month): Single-user access for researchers, students, or technicians analyzing cement microscopy images.

Team Plan (\$30/month): Multi-user access for small research teams or laboratories (up to 3 users).

Institutional Plan (\$40/month): Designed for universities, large laboratories, or cement factories, allowing class or group registrations (up to 20 users).

This tiered subscription model ensures affordability for individual learners while also supporting scalability for research institutions and industry applications.

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9. APPENDICIES

Appendix – A : Plagiarism

